



**SRM INSTITUTE OF SCIENCE AND TECHNOLOGY  
COLLEGE OF ENGINEERING AND TECHNOLOGY  
DEPARTMENT OF NETWORKING AND COMMUNICATIONS  
18CSP109L / 18CSP111L & 18CSP112L - MAJOR PROJECT /  
MAJOR PROJECT & INTERNSHIP**

# **Generative AI chatbot for Student Counselling**

**Udit Kumar Reg No: RA2011031010057**

**Priyanshu Maheshwari Reg No: RA2011028010091**

**Batch ID: NWC150057**

**Guide name and Designation: Dr. A. Helen Victoria (Assistant  
Professor)**



**SRM**  
INSTITUTE OF SCIENCE & TECHNOLOGY  
Deemed to be University u/s 3 of UGC Act, 1956

# Table of contents

- Introduction
- Problem statement
- Objectives
- Literature Survey
- Architecture Diagram
- Algorithm

# Introduction

Generative AI refers to a category of artificial intelligence systems that are designed to generate new content or data that resembles human-generated content. Unlike traditional AI systems that are programmed to follow predefined rules or patterns, generative AI models can create original and novel outputs based on patterns and examples learned from large datasets

# Problem Statement

In an era where students face an ever-increasing array of academic and personal challenges, our **Generative AI Chatbot** is here to bridge the gap between the demand for counseling services and the limited availability of human counselors. With its ability to understand, empathize, and respond to students' concerns in real-time, this chatbot promises to revolutionize the way students seek guidance and support throughout their educational journey.

# Objectives

- The need for student counseling arises from various factors that impact students' academic performance, mental well-being, and overall personal growth. Some key reasons for the need for student counseling include: academic support , emotional and mental health , career guidance and personal development
- **Anonymous and Non-Judgmental Environment:** Many students may feel more comfortable discussing sensitive or personal issues with a chatbot rather than a human counselor. The anonymous and non-judgmental nature of the chatbot can encourage open communication.
- **Prompt Responses:** The chatbot can provide immediate responses to students, reducing waiting times and addressing their concerns promptly.
- **24/7 Availability:** The chatbot can be available round-the-clock, allowing students to seek counseling support whenever they need it, even outside regular working hours or during holidays.



# literature survey

| Year/Journal/Indexing | Author/Title of the Paper                                                                          | Techniques                                      | Action/Findings                                               |
|-----------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------|---------------------------------------------------------------|
| 2017 - ACL            | Rajpurkar et al. - "SQuAD: 100,000+ Questions for Machine Comprehension of Text"                   | Question-Answering Models, Attention Mechanisms | Introduced SQuAD dataset, a benchmark for contextual QA.      |
| 2018 - ACL            | Devlin et al. - "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding" | Pre-training Transformer-based models           | BERT achieved state-of-the-art results on multiple NLP tasks. |

|                |                                                                                                                                    |                                                         |                                                           |
|----------------|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|-----------------------------------------------------------|
| 2020 - NeurIPS | Lewis et al. - "BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension" | Sequence-to-Sequence Pre-training, Denoising Objective  | BART showed strong performance on various language tasks. |
| 2019 - ACL     | Yang et al. - "XLNet: Generalized Autoregressive Pretraining for Language Understanding"                                           | Autoregressive Pre-training, Permutation-based Training | XLNet surpassed BERT in several NLP benchmarks.           |

|                          |                                                                        |                                                               |                                                             |
|--------------------------|------------------------------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------|
| 2021 - ACL               | Lewis et al. - "DPR: A Simple Way to Start Conversational Search"      | Dense Passage Retrieval (DPR), Retrieval-Augmented Generation | DPR improved conversational search using a dense retriever. |
| 2022 - IEEE Trans. on AI | Zhang et al. - "Reinforcement Learning for Dialogue Systems: A Review" | Reinforcement Learning, Dialogue Systems                      | Reviewed RL approaches for dialogue systems.                |



|              |                                                                                              |                                                          |                                                                                        |
|--------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------|
| 2020 - arXiv | Vaswani et al. -<br>"Attention Is All You Need"                                              | Transformer<br>Architecture, Self-<br>Attention          | Introduced the<br>Transformer model<br>architecture.                                   |
| 2022 - IJCAI | Wang et al. -<br>"Dialogue<br>Contextualization for<br>Open-domain<br>Conversational Agents" | Contextualization, Pre-<br>training, Dialogue<br>Systems | Proposed a method for<br>improving<br>conversational agents'<br>context understanding. |

# Drawback

## The Power and Limitations of Large Language Models

- LLMs are trained on huge amounts of text with the aim of learning the conditional distribution of words in a language. Doing so allows them to generalize and generate meaningful text without directly memorizing the training data. This means they can accurately recall widely disseminated information, such as historical events or popular cultural facts.
- However, the LLM's knowledge is restricted to its training set. So, suppose the model was trained on data up to 2021 and is asked about a company founded in 2023. In that case, it may generate a plausible but entirely fabricated description - a phenomenon known as "**hallucination**." Managing hallucinations is tricky, especially in applications where accuracy and reliability are paramount, such as customer-service chatbots, knowledge-base assistants, or AI tutors.

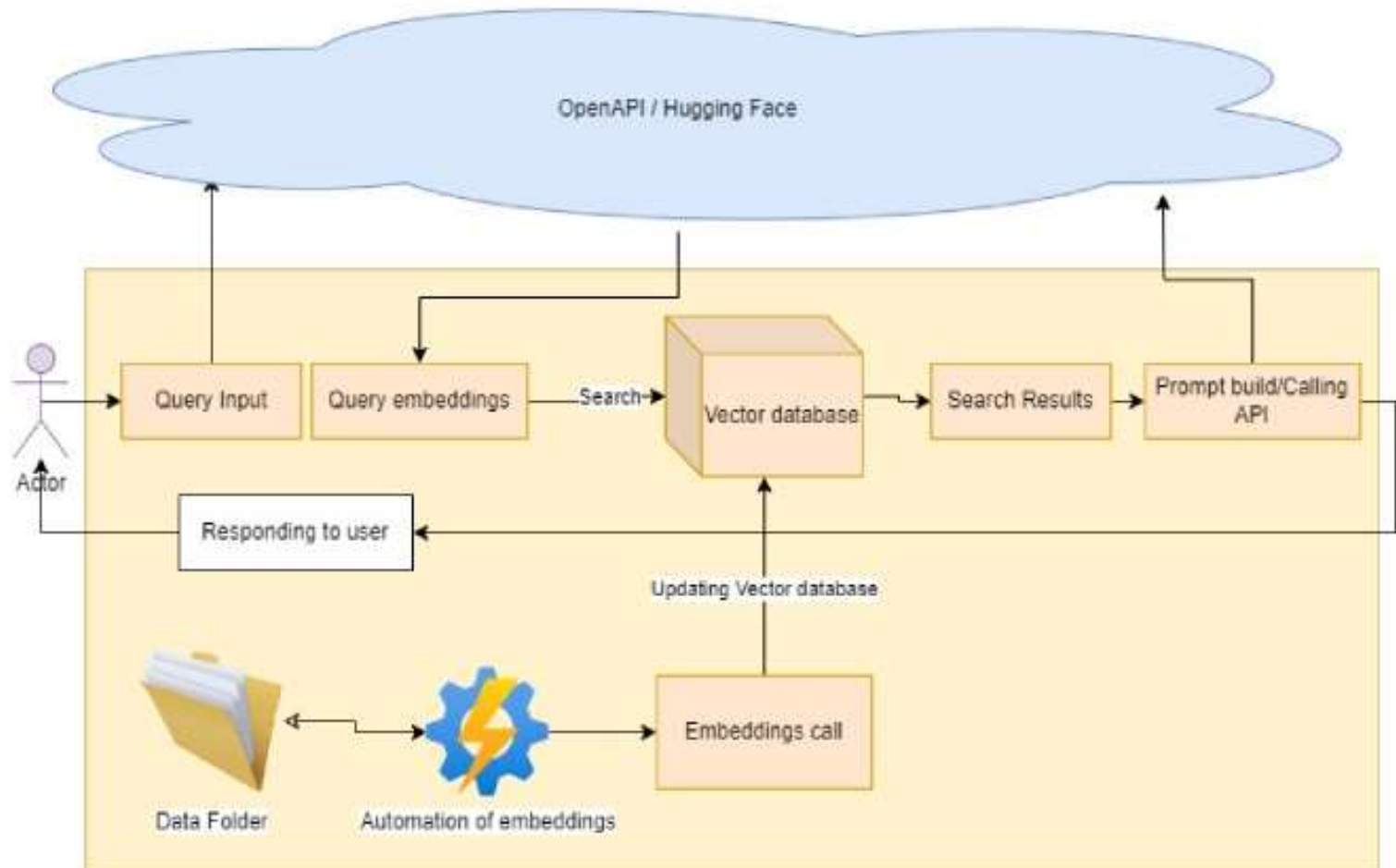
- One promising strategy to mitigate hallucination is the **use of retrievers** in tandem with LLMs. A retriever fetches relevant information from a trusted knowledge base (like a search engine), and the LLM is then specifically prompted to rearrange the information without inventing additional details.
- LLMs' large context window sizes facilitate the inclusion of multiple documents in a single prompt. Models like GPT-4 and Claude can handle context windows of up to 32k and 100k tokens, respectively, equating to approximately 20k words or 40 pages of text. However, the cost of execution rises with the number of tokens used, hence the need for an efficient retriever to find the most relevant documents.



**SRM**

INSTITUTE OF SCIENCE & TECHNOLOGY  
Deemed to be University u/s 3 of UGC Act, 1956

# Architecture Diagram





**SRM**  
INSTITUTE OF SCIENCE & TECHNOLOGY  
Deemed to be University u/s 3 of UGC Act, 1956

# Algorithm/Steps

## **1. Initialization:**

1. Load the pre-trained Generative AI Model (e.g., GPT-3 or GPT-4).
2. Set up the NLP preprocessing module.
3. Initialize the User Data Storage, Counselor Knowledge Base, and other necessary components.

## **2. User Interaction:**

1. User interacts with the chatbot through a UI.
2. Chatbot receives user input.

## **3. NLP Processing:**

1. Preprocess user input to understand intent, entities, and context.
2. Extract keywords and relevant information.



#### **4. Generate Response:**

1. Utilize the Generative AI Model to generate a natural language response.
2. Incorporate extracted keywords and context into the response.

#### **5. User Data Management:**

1. Store user interactions, including questions and chatbot responses, in the User Data Storage.
2. Maintain context for the ongoing conversation.

#### **6. Access Counselor Knowledge Base:**

1. Retrieve relevant counseling resources from the Counselor Knowledge Base if necessary.
2. Share resources and information with the user.

**7. Privacy and Security:**

1. Ensure user data is stored securely, following privacy regulations.
2. Authenticate users when necessary for privacy and security.

**8. Feedback and Learning:**

1. Collect user feedback on the chatbot's responses.
2. Continuously learn and improve responses based on user feedback.

**9. Integration:**

1. Interface with external systems and databases (Integration APIs) to access student information, schedules, or other relevant data when required.

**10. Analytics and Monitoring:**

1. Monitor chatbot performance, user satisfaction, and system health.
2. Use analytics to optimize responses and improve user experience.

## **11. Notification and Alerting:**

1. Trigger notifications to human counselors when user inquiries require human intervention or follow-up.

## **12. Loop:**

1. Repeat steps 2-11 for each user interaction.

## **13. Exit:**

1. Allow users to end the conversation or seek further assistance as needed.



The algorithm described to build the generative AI chatbot for student counseling is a combination of various techniques and components rather than a single, specific algorithm with a unique name. It draws from the fields of Natural Language Processing (NLP), Natural Language Generation (NLG), and conversational AI, utilizing pre-trained language models like GPT (Generative Pre-trained Transformer) for text generation and understanding.



**SRM**  
INSTITUTE OF SCIENCE & TECHNOLOGY  
*Deemed to be University u/s 3 of UGC Act, 1956*

Thanks